DS 5030
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Jack Brennan and Austin Blackburn
Lab 6 Report

This report gives a review of a randomized controlled trial comparing three interventions intended to prevent progression from pre-diabetes to type 2 diabetes (metformin (met), rosiglitazone (rosi), and structured lifestyle changes (diet and exercise)). The dataset contains 699 participants randomly assigned to one of three groups: 232 to met, 233 to rosi, and 234 to lifestyle. A participant is recorded as a success if they did not progress to diabetes during the trial, and as a failure if they did. The arms should not differ systematically at baseline due to being randomly sampled. The difference in success rates between any two arms is an unbiased estimate of the average treatment effect (ATE) between those interventions. To quantify the uncertainty in these estimates, we bootstrapped the outcomes 10,000 times. Each time resampled participants with replacement within each arm. For each of the three comparisons (rosi vs. met, rosi vs. lifestyle, and met vs. lifestyle), we computed bootstrap confidence intervals at the 90%, 95%, and 99% levels and treated a comparison as statistically significant when its interval excluded zero.

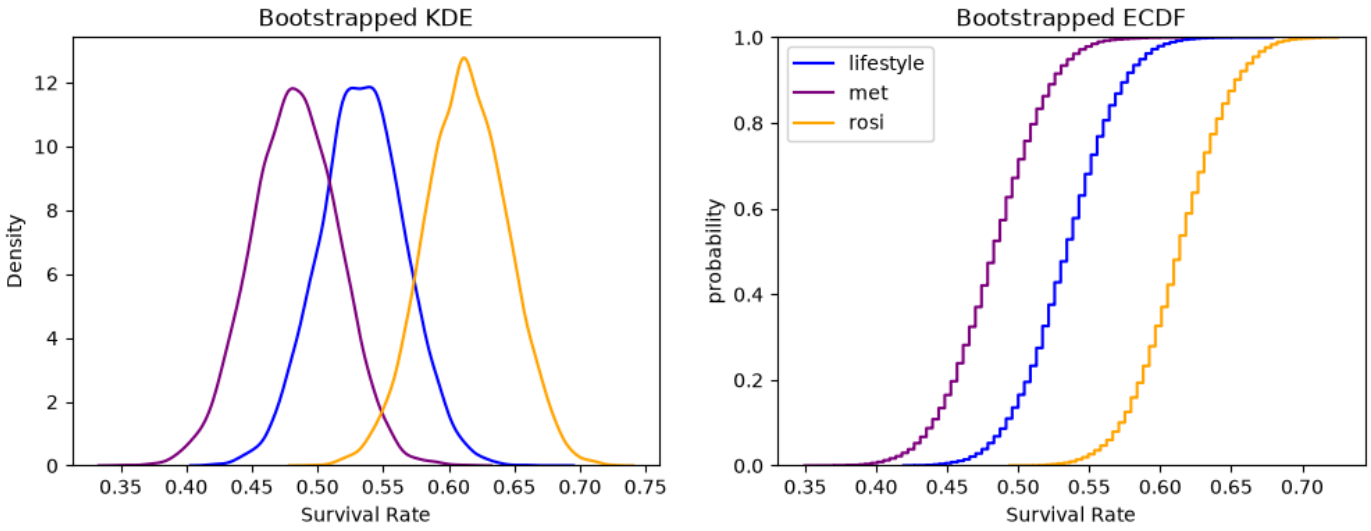
Table 1. Observed outcomes by treatment arm

Arm	n	Successes	Failures	Success rate
Rosiglitazone	233	143	90	61.4%
Lifestyle	234	125	109	53.4%
Metformin	232	112	120	48.3%

Table 2. Pairwise average treatment effects (percentage points), 10,000 bootstrap resamples

Comparison	ATE	95% CI	Excludes zero at
Met − Rosi	−13.1	[−22.1, −4.1]	90%, 95%, 99%
Rosi − Lifestyle	+8.0	[−1.0, 16.9]	none
Met − Lifestyle	−5.1	[−14.2, 3.9]	none

Figure 1:



Success rates differed across the three arms, as shown in Table 1. Rosiglitazone had the highest rate of non-progression (61.4%), then lifestyle modification (53.4%) and then metformin (48.3%). Figure 1 shows the bootstrap distributions of the arm-level success rates. The rosiglitazone and metformin distributions overlap very little, whereas lifestyle is between them and overlaps with both by a lot. The comparisons in Table 2 indicate that only one difference is statistically significant. Rosiglitazone outperformed metformin by an estimated 13.1%, and the interval excluded zero at all three confidence levels. Rosiglitazone's estimated 8.0% advantage over lifestyle change was not statistically significant at any level tested and the 5.1% difference between lifestyle change and metformin was also not statistically significant.

In conclusion, the trial provides evidence in favor of rosiglitazone relative to metformin, but it does not establish a clear best intervention overall. Although rosiglitazone had the highest success rate and was the only intervention that was statistically significant relative to another arm, its advantage was over metformin only, not over lifestyle modification. Its success relative to lifestyle modification was not statistically significant and lifestyle modification could not be distinguished from either pharmacological treatment, so given this it remains a good baseline approach. Metformin had the lowest success rate and was outperformed by rosiglitazone, and did not show any evidence of a benefit over lifestyle change. One major limitation in these results is

that the dataset records only progression to diabetes and contains no information on external data, cost, or long-term health effects of these options, which may influence a regulatory decision. Based only on the evidence from the trial, we would recommend rosiglitazone over metformin, but neither of the treatments over lifestyle modification.